

August 12, 2024

Energy Division  
California Public Utilities Commission  
505 Van Ness Avenue  
San Francisco, California 94102  
EDTariffUnit@cpuc.ca.gov

**RE: Comments of the Vehicle Grid Integration Council on Draft Resolution E-5313**

Dear Sir or Madam:

Pursuant to the provisions of General Order 96-B and Comment Letter E-5313 circulated on July 22, 2024, the Vehicle-Grid Integration Council (“VGIC”) hereby submits these comments to Draft Resolution E-5315, which approves, with modifications the Advice Letter 5185-E of Southern California Edison (“SCE”) Company, Advice Letter 7125-E of Pacific Gas and Electric Company, (“PG&E”), and Advice Letter 4350-E of San Diego Gas and Electric Company (“SDG&E”), Lessons Learned from the Vehicle to Grid Alternating Current Pilot and Recommendations for A Pathway Forward (“Joint Advice Letter”) submitted on January 5, 2024.

Overall, VGIC continues to support implementing the temporary pathway for vehicle-to-grid (“V2G”) alternating current (“AC”) interconnection. Therefore, VGIC submitted a response supporting the Joint Advice Letter to extend the V2G AC pilots. The Draft Resolution correctly determines that the pilot should be extended and that alternative interim interconnection pathways should be considered to allow for additional pilot participation. However, in response to the Draft Resolution, VGIC offers several key areas of concern in the comments below that must be addressed in the final adopted resolution.

**I. VGIC SUPPORTS THE APPROVAL OF THE EXTENSION OF THE V2G AC PILOT.**

VGIC supports the Commission’s determination that the IOUs should “reopen the V2G AC Interconnection Pilot and extend it until tariff modifications for a permanent interconnection process for V2G AC systems are approved by the Commission.” As highlighted in the Draft Resolution, extending the pilot pathway will allow for additional learnings on V2G AC interconnection, which is still a nascent interconnection pathway that various industry stakeholders are still developing and further understanding. These learnings will be important for the further development of a more permanent interconnection pathway in Rule 21.

While the original Joint Advice Letter requested an extension of the pilots only through 2025, the Commission is correct in determining that there should not be a definitive calendar date for the sunset of these pilots. VGIC agrees that given the changing and, at times, lengthy timelines of standards approval, it makes sense to maintain flexibility in the timeline for the extension of the pilot in order to “avoid gaps in the interconnection process for V2G AC pilot projects between the end of the V2G AC Interconnection Pilot and the time a permanent interconnection process for V2G AC systems is added to, and becomes effective in, Rule 21.”<sup>1</sup> This will also avoid having the IOUs submit any future non-substantive Advice Letters to merely request further extensions of the pilots because of delays.

## **II. THE COMMISSION SHOULD CORRECT ERRORS RELATED TO THE PROCESS FOR ALLOWING ADDITIONAL INTERIM APPROACHES.**

VGIC also supports the Draft Resolution approving the IOUs’ ability to allow additional interim V2G AC pathways, including those not requiring a relay, to be used in this pilot. The current interim V2G AC interconnection pathway’s requirement of relays is likely reducing the number of participants seeking to interconnect under this pathway, given the significant cost of relay devices. There may be other approaches that could be used to provide alternative interim interconnection pathways, and VGIC agrees with the Draft Resolution that the IOUs should be allowed to pursue those pathways, which will undoubtedly offer valuable insights to shape the long-term initiative to incorporate V2G AC in Rule 21.

However, process outlined in the Draft Resolution to approve alternative pathways includes an industry-led working group and Advice Letter process. This is misaligned with the steps needed to support an alternative interim pathway. The Draft Resolution outlines roughly 3 months of “industry-led” working groups followed by the submission of a Tier 2 Advice Letter roughly a month later. While staff approval of a Tier 2 Advice Letter is faster than the process of Resolution approval, it will likely take several months for the Advice Letter to be approved. This means that customers will not be able to utilize the alternative interim pathway for at least 6 months, in the best case scenario.

Moreover, an alternative interim pathway cannot emerge from stakeholder discussion and Advice Letters alone. Demonstration, piloting, field testing and validation, or other real-world experience is required to establish an alternative interim pathway. Put differently, the Draft Resolution’s proposed process is ordered backwards: the industry-led work, informed by the IOUs, and subsequent Advice Letter process could not progress without first developing real-world experience. This has been made abundantly clear from the development of UL 1741 SC, where the core specifications that comprise the standard emerged from real-world experience in SCE’s EPIC V2G AC Pilot. While the UL 1741 SC committee is diligently working through a

---

<sup>1</sup> Draft Resolution at 8

critical consensus-driven stakeholder process, its origins are in real-world experience led by SCE. VGIC expects this same dynamic – experimenting *first* – would hold true for an interim non-relay pathway, as well.

With this in mind, VGIC strongly urges the Commission to allow the IOUs to gain experience with interim approaches by working directly with pilot participants to allow interconnection without the use of a relay. While VGIC understands the value of establishing a consensus-driven interim approach that “must be applicable industry wide,”<sup>2</sup> this must be balanced with the need to efficiently establish such interim approaches and acknowledgement that they cannot emerge without real-world piloting. VGIC believes that without this flexibility, IOUs will have no means to experiment with applicable interim pathways that could inform the consensus-driven, industry-wide approach envisioned by the Draft Resolution. The Commission should aim to further the effectiveness of this pilot as a space to provide valuable data on real-life V2G AC interconnection that can be used to determine the best way to advance V2G AC given the current standards environment.

### **III. THE RESOLUTION SHOULD NOT FORECLOSE THE POTENTIAL FOR INTERIM INTERCONNECTION PATHWAYS TO BECOME PERMANENT.**

In the Draft Resolution, the Commission cites the Joint Advice Letter, stating that “Once standard and certification requirements for V2G-AC have been published and Rule 21 is updated, all new V2G-AC projects must comply with those requirements and *the alternative certification process developed for V2G-AC pilot projects will no longer be available for use.*”<sup>3</sup> VGIC believes that the intent behind this phrasing is to state that once a methodology for V2G AC interconnection is adopted under Rule 21, customers seeking interconnection of these devices must follow Rule 21 protocols, instead of using the non-Rule 21 interim pilot pathway. However, the Commission should not automatically assume that interim pathways from the pilot will not be integrated into Rule 21 as permanent pathways for interconnection. There are many stakeholders anticipating the release of UL 1741 SC, and it is implied in the Joint Advice Letter and Draft Resolution that this UL standard will form the basis for a permanent Rule 21 pathway. However, this is not a foregone conclusion. The IOUs and pilot participants may find that interim pilot solutions could be found as acceptable interconnection standards that could be integrated into Rule 21. It is also very possible that multiple pathways could be integrated into Rule 21 to ensure safe and reliable interconnection while also supporting customer choice. This principle of customer choice is not uncommon within Rule 21, which holds several options and pathways for any given technology to move through the Rule 21 process.

---

<sup>2</sup> Draft Resolution at 9

<sup>3</sup> Draft Res at 9 citing Joint AL at 4

Lastly, VGIC would like to note that customers that have interconnected under a pilot pathway should be allowed to continue to operate under that pathway if they so choose. Customers should not have to re-apply for interconnection under Rule 21 once their system is operational.

#### **IV. CONCLUSION.**

VGIC appreciates the opportunity to submit these comments on the Draft Resolution. We look forward to further collaboration with the Commission and stakeholders on this initiative.

Respectfully submitted,

/s/ Zach Woogen

Zach Woogen

Interim Executive Director

**VEHICLE-GRID INTEGRATION  
COUNCIL**

cc: Jose Aliaga-Caro, CPUC ([Jose.Aliaga-Caro@cpuc.ca.gov](mailto:Jose.Aliaga-Caro@cpuc.ca.gov))  
Caitlin Pollock, CPUC ([Caitlin.Pollock@cpuc.ca.gov](mailto:Caitlin.Pollock@cpuc.ca.gov))  
Connor Flanigan, SCE ([AdviceTariffManager@sce.com](mailto:AdviceTariffManager@sce.com))  
Adam Smith, SCE, c/o Karyn Gansecki ([Karyn.Gansecki@sce.com](mailto:Karyn.Gansecki@sce.com))  
Sidney Bob Dietz II, PG&E, c/o Megan Lawson ([PGETariffs@pge.com](mailto:PGETariffs@pge.com))  
Greg Anderson, SDG&E ([GAnderson@sdge.com](mailto:GAnderson@sdge.com), [SDGETariffs@sdge.com](mailto:SDGETariffs@sdge.com))  
Service list R.17-07-007  
Service list R.18-12-006